CYBER SECURITY

BCE 8028

Year: IV **Semester: VIII**

Teaching Schedule Hours/Week					Examination Schedule						Total Marks
					Final				Internal Assessment		
					Theory		Practical		Theory Marks	Practical Marks	
Credit Hours	L	T	P	Total	Duration	Marks	Duration	Marks	40	25	125
3	3	1	2	6	3 Hrs.	60	-	-			

Note: **L:** Lecture **T:** Tutorial **P:** Practical

Course Objectives:

- To understand cyber threats, vulnerabilities, and attacks
- To learn security mechanisms for networks, systems, and applications
- To apply cryptographic techniques for secure communication
- To understand cyber laws, ethics, and compliance
- To build awareness of emerging cyber security challenges

Course Contents:

Unit 1: Introduction to Cyber Security (7 Hours)

- Introduction to Cyber Space and Cyber Security
- CIA Triad: Confidentiality, Integrity, Availability
- Cyber Security terminology and taxonomy
- Threats, vulnerabilities, risks, and attacks
- Types of attackers: hackers, crackers, insiders, script kiddies
- Cyber kill chain
- Security policies, standards, and best practices

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गोठगाउँ, मोरङ, नेपाल
1995
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- Overview of security models (Bell-LaPadula, Biba Model)

Unit 2: Cyber Threats and Attack Techniques (8 Hours)

- Malware: viruses, worms, trojans, spyware, ransomware
- Phishing, spear phishing, whaling attacks
- Password attacks: brute force, dictionary, rainbow tables
- Social engineering attacks
- Web-based attacks: SQL injection, XSS, CSRF
- Denial of Service (DoS) and Distributed DoS (DDoS) attacks
- Insider threats
- Case studies of major cyber attacks

Unit 3: Cryptography and Network Security (10 Hours)

3.1 Cryptography:

- Introduction to cryptography and cryptanalysis
- Symmetric key cryptography (DES, AES)
- Asymmetric key cryptography (RSA, Diffie-Hellman, ECC)
- Hash functions (MD5, SHA family)
- Digital signatures and certificates
- Public Key Infrastructure (PKI)

3.2 Network Security

- Secure communication protocols: SSL/TLS, HTTPS
- IP Security (IPSec)
- Firewalls: types and configurations
- Virtual Private Networks (VPNs)

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Unit 4: System, Application, and Cloud Security (10 Hours)

4.1 System Security

- Operating system security concepts
- Authentication and authorization mechanisms
- Access control models (DAC, MAC, RBAC)
- Secure boot and trusted computing

4.2 Application Security

- Secure software development lifecycle (SSDLC)
- OWASP Top 10 vulnerabilities
- Code injection and buffer overflow attacks
- Secure coding practices

4.3 Cloud & IoT Security

- Cloud security challenges and threats
- Cloud service models and security (IaaS, PaaS, SaaS)
- IoT security issues and countermeasures

Unit 5: Cyber Laws, Digital Forensics & Emerging Trends (10 Hours)

5.1 Cyber Laws & Ethics

- Cyber ethics and professional responsibilities
- Nepal IT Policy 2015 (i.e. Nepal IT Policy 2072)
- Cyber crimes and legal procedures
- Intellectual property rights in cyberspace
- Privacy and data protection laws (GDPR overview)

5.2 Digital Forensics

- Introduction to digital forensics
- Evidence collection and preservation
- Network and system forensics

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- Incident response lifecycle

5.3 Emerging Trends

- Artificial Intelligence in Cyber Security
- Blockchain and security
- Zero Trust Architecture
- Future challenges in cyber security

Text Books

1. **William Stallings**, *Cryptography and Network Security*, Pearson
2. **Nina Godbole & Sunit Belapure**, *Cyber Security*, Wiley
3. **Charles P. Pfleeger**, *Security in Computing*, Pearson

Reference Books

- Eric Cole, *Network Security Bible*, Wiley
- Thomas R. Peltier, *Information Security Policies, Procedures, and Standards*, Auerbach
- OWASP Documentation (latest)

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